Supervised and unsupervised WSD based on a Naïve Bayes Model

- **In order to formalize the described model, we shall present the probability structure of the corpus C.**

- **Notations:**

  w **– target word (word to be disambiguated)**
  $s_1, \ldots, s_k$ **- possible senses for** w
  $c_1, \ldots, c_I$ **- contexts of** w **in a corpus** C
  $v_1, \ldots, v_J$ **- words used as contextual features for the disambiguation of** w

- **The probability structure of the corpus is based on one main assumption:** ***the contexts {ci, i} in the corpus C are independent*****. Hence, the likelihood of *C* is given by the product**

$$P(C) = \prod_{i=1}^{I} P(c_i)$$

On considering the possible senses of each context, one gets

$$P(C) = \prod_{i=1}^{I} \sum_{k=1}^{K} P(s_k) \cdot P(c_i \mid s_k)$$

- **A model with independent features (usually known as the Naïve Bayes model) assumes that the contextual features are conditionally independent. That is,**

$$P(c_i \mid s_k) = \prod_{v_j \ in \ c_i} P(v_j \mid s_k) = \prod_{j=1}^{J} (P(v_j \mid s_k))^{|v_j \ in \ c_i|},$$

where by $|v_j$ in $c_i|$ **we denote the number of occurrences of feature** v_j **in context** c_i**. Then, the likelihood of the corpus** C **is**

$$P(C) = \prod_{i=1}^{I} \sum_{k=1}^{K} P(s_k) \prod_{j=1}^{J} (P(v_j \mid s_k))^{|v_j \ in \ c_i|}.$$

- **<u>The parameters</u> of the probability model with independent features are**

$$\{P(s_k),\ k = 1,\dots,K \text{ and } P(v_j \mid s_k),\ j = 1,\dots,J,\ k = 1,\dots,K\}.$$

- **<u>Notation:</u>**

$P(s_k) = \alpha_k,\ k = 1,\dots,K,\ \alpha_k \geq 0$ **for all** k, $\sum_{k=1}^{K} \alpha_k = 1$

$P(v_j \mid s_k) = \theta_{kj},\ k = 1,\dots,K,\ j = 1,\dots,J,\ \theta_{kj} \geq 0$ **for all** k **and** j,

$\sum_{j=1}^{J} \theta_{kj} = 1$ **for all** $k = 1, \dots, K$

With this notation, the likelihood of the corpus C can be written as

$$P(C) = \prod_{i=1}^{I} \sum_{k=1}^{K} \alpha_k \prod_{j=1}^{J} (\theta_{kj})^{|v_j \ in \ c_i|}.$$

The well known Bayes classifier involves the a posteriori probabilities of the senses, calculated by the Bayes formula for a specified context c,

$$P(s_k \mid c) = \frac{P(s_k) \cdot P(c \mid s_k)}{\sum_{k=1}^{K} P(s_k) \cdot P(c \mid s_k)} = \frac{P(s_k) \cdot P(c \mid s_k)}{P(c)}$$

with the denominator independent of senses. The Bayes classifier chooses the sense s' for which the a posteriori probability is maximal (sometimes called the Maximum A Posteriori classifier)

$$s' = \arg\max_{k=1,\ldots,K} P(s_k \mid c).$$

Taking into account the previous Bayes formula, one can define the Bayes classifier by the equivalent formula

$$s' = \arg\max_{k=1,\ldots,K} (\log P(s_k) + \log P(c \mid s_k)).$$

Of course, when implementing a Bayes classifier, one has to estimate the parameters first.

Parameter estimation

Parameter estimation is performed by the <u>Maximum Likelihood method</u>, for the available corpus *C*. That is, one has to solve the optimization problem

$$\max\left(\log P(C) \mid \{P(s_k),\ k=1,\ldots,K \ \text{ and } \ P(v_j \mid s_k),\ j=1,\ldots,J,\ k=1,\ldots,K\}\right).$$

For the Naïve Bayes model, the problem can be written as

$$\max\left(\sum_{i=1}^{I}\log\left(\sum_{k=1}^{K}\alpha_k\prod_{j=1}^{J}(\theta_{kj})^{|v_j \text{ in } c_i|}\right)\right) \tag{1}$$

with the constraints

$$\sum_{k=1}^{K}\alpha_k = 1,\ \sum_{j=1}^{J}\theta_{kj} = 1 \text{ for all } k = 1, \ldots, K$$

For *<u>supervised disambiguation</u>*, where an annotated training corpus is available, the parameters are simply estimated by the corresponding frequencies (as in last week's lecture):

$$\hat{\theta}_{kj} = \frac{|\text{occurrences of } v_j \text{ in a context of sense } s_k|}{\sum_{j=1}^{J}|\text{occurrences of } v_j \text{ in a context of sense } s_k|},\ k = 1, \ldots, K;\ j = 1, \ldots, J$$

$$\hat{\alpha}_k = \frac{|\text{occurrences of sense } s_k \text{ in } C|}{|\text{occurrences of } w \text{ in } C|},\ k = 1, \ldots, K$$

For *unsupervised disambiguation*, where no annotated training corpus is available, the maximum likelihood estimates of the parameters are constructed by means of the Expectation-Maximization (EM) algorithm.

- **For the unsupervised case, the optimization problem (1) can be solved only by iterative methods. The Expectation Maximization algorithm is a very successful iterative method, known as very well fitted for models with missing data. (Here the missing data are the senses of the ambiguous word).**

- **Each iteration of the algorithm involves two steps:**
  - ✓ **estimation of the missing data by the conditional expectation method (E-step)**
  - ✓ **estimation of the parameters by maximization of the likelihood function for complete data (M-step)**

  The E-step calculates the conditional expectations given the current parameter values, and the M-step produces new, more precise parameter values. The two steps alternate until the parameter estimates in iteration $r + 1$ and r differ by less than a threshold ε.

- **The EM algorithm is guaranteed to increase the likelihood $\log P(C)$ in each step. Therefore, two stopping criteria for the algorithm could be considered:**
  - **Stop when the likelihood $\log P(C)$ is no longer increasing significantly;**
  - **Stop when parameter estimates in two consecutive iterations no longer differ significantly.**

- **The available data, called *incomplete data*, are given by the corpus C. The *missing data* are the senses of the ambiguous words, hence they must be modeled by some random variables. The *complete data* consist of incomplete and missing data.**

Unsupervised WSD with the Naïve Bayes model – Feature selection

- **The classical clustering technique is represented by the Naïve Bayes model trained with the EM algorithm.**
- **When the Naïve Bayes model is applied to supervised disambiguation, the actual words occurring in the context window are usually used as features. This type of framework generates a great number of features and, implicitly, a great number of parameters. This can dramatically decrease the model performance since the available data is usually insufficient for the estimation of the great number of resulting parameters. This situation becomes even more drastic in the case of unsupervised disambiguation, where parameters must be estimated in the presence of missing data (the sense labels).**
- **In order to overcome this problem, the various existing unsupervised approaches to WSD implicitly or explicitly perform a feature selection.**

Semantic WN-based feature selection

- **This approach is based on a set of features formed by the actual words occurring near the target word (within the context window) and reduces the size of this feature set by performing knowledge-based feature selection that relies entirely on WN.**

- **The WN semantic network provides the words considered as relevant for the set of senses taken into consideration corresponding to the target.**

- **The value of a feature is given by the number of occurrences of the corresponding word in the given context window.**

Semantic WN - based Feature Selection

- **We start by choosing words occurring in the same WN synsets as the target word (WN synonyms), corresponding to all senses of the target. Additionally, the words occurring in the synsets related (through explicit relations provided in WN) to those containing the target word are also considered as part of the so-called "disambiguation vocabulary". Synsets and relations are restricted to those associated with the part of speech of the target. The content words of the glosses and the example strings of all types of synsets participating in the disambiguation process are equally considered. Usage of the example string as well is in accordance with previous studies performed for knowledge-based WSD.**

Syntactic Dependency-based Feature Selection

- This type of features augments the role of linguistic knowledge.

- Syntactic features are provided by dependency relations as defined by the classical Dependency Grammar formalism.

- **The semantic space that is proposed to the Naïve Bayes model for unsupervised WSD is entirely based on syntactic knowledge, more precisely on dependency relations, extracted from natural language texts via a syntactic parser. The parser extracts dependency relations that will indicate the disambiguation vocabulary required by Naïve Bayes. This is how the disamb. voc. is formed:**

- **It is various combinations of dependencies, pointed out by the obtained <u>dependency structure</u>, that form the <u>context</u> over which the semantic space for WSD is constructed. <u>The number of features</u> (coming from all resulted dependencies) <u>is decreased</u> by taking into account only specific dependency relations. The <u>disambiguation vocabulary</u> is formed by considering all words that participate in the considered dependencies.**

Syntactic Dependency-based Feature Selection

In order to inform the construction of the semantic space for WSD with syntactic knowledge of this type, several elements must be taken into consideration. At the first stage of the experiments, we make no qualitative distinction between the different relations, by not taking into account the type of the involved dependencies. At the second stage of the experiment, we can inform the construction of the semantic space in a more linguistically rich manner, by considering the dependency type. Therefore we make use only of specific dependency relations, which are considered more informative than others relatively to the studied part of speech.

Syntactic Dependency-based Feature Selection

- **In defining the syntactic context of the target word, we can first take into consideration direct relationships (first order dependencies) between the target and other words, where the target can be either the head or a dependent, and which correspond to paths of length 1 anchored at the target in the associated dependency graph. At the next stage we can consider indirect relationships between the target and other words, by taking into account paths of length 2 (second order dependencies) in the same associated graph. The order of these dependencies (lengths of the paths in the associated graph) represents a parameter that can vary and which can be considered analogous to the classical "window size" parameter. Just as in the case of the window size, this parameter should have relatively small values, since it is a known fact that linguistically interesting paths are of limited length. This justifies the usual choice of limiting the investigation to second order dependencies.**

- **Separate experiments, which view the <u>target word</u> as <u>head</u> and as <u>dependent</u>, respectively, can be carried out both corresponding to first order and to second order dependencies.**

Syntactic Dependency-based Feature Selection

- **When using the Naïve Bayes model as clustering technique, it is observed that a small number of dependency types should be considered, if possible just one, in order to decrease the number of parameters that must be estimated by the EM algorithm.**

  This conclusion would not hold in the case of a clustering technique requiring a high number of features (such as spectral clustering, for instance). In such a case, several types of dependencies would probably have to be considered, which would create a more linguistically informed semantic space for WSD – that could be even more beneficial.

Syntactic Dependency-based Feature Selection

While the classical dependency-based linguistic theory does not allow the arches denoting the dependency relations to intersect (thus leading to an oriented graph which has no cycles), the dependency analysis performed by the Stanford parser, for instance, can be either projective (disallowing crossing dependencies) or non-projective (permitting crossing dependencies). The number of resulting features is then decreased by taking into account only dependency relations of specific types. Both types of analyses (non-projective and projective) have been performed and discussed in the literature, relatively to the presented case studies, with the projective syntactical analysis leading to the best results so far.